NOVA SCHOOL OF
SCIENCE & TECHNOLOGY

# STR

## TP3 – LAB#1

### Petri Nets

2022 - 2023

# TOPICS

## PART 1:
## Software installation and development of first PN models

- Revisiting Petri net Modeling
- Installing HPSim software
- Implementation of examples from theoretical classes
- Development of several modules to training on PN concepts

## PART 2:
## DLL integration in HPSim software

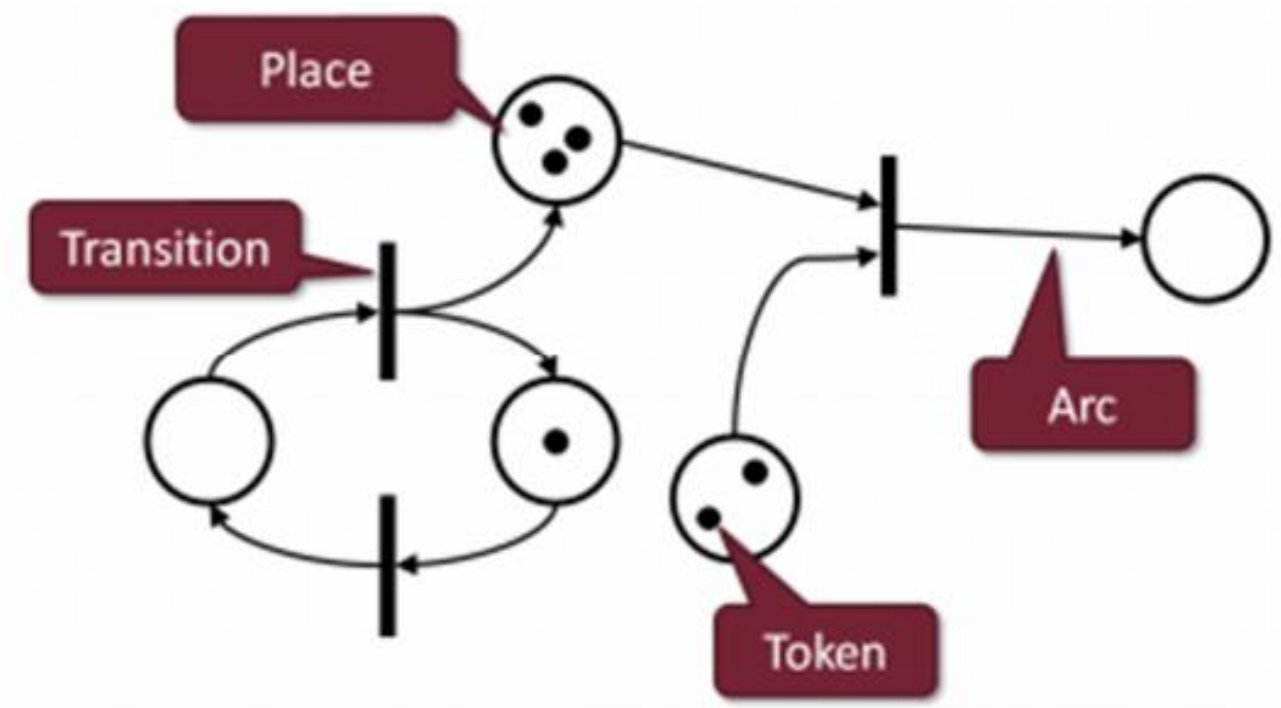
- Installing new version of HPSim software
- Creating a DLL
- Using the DLL in HPSim
- First examples

# Part 1

Software installation and development of first PN models

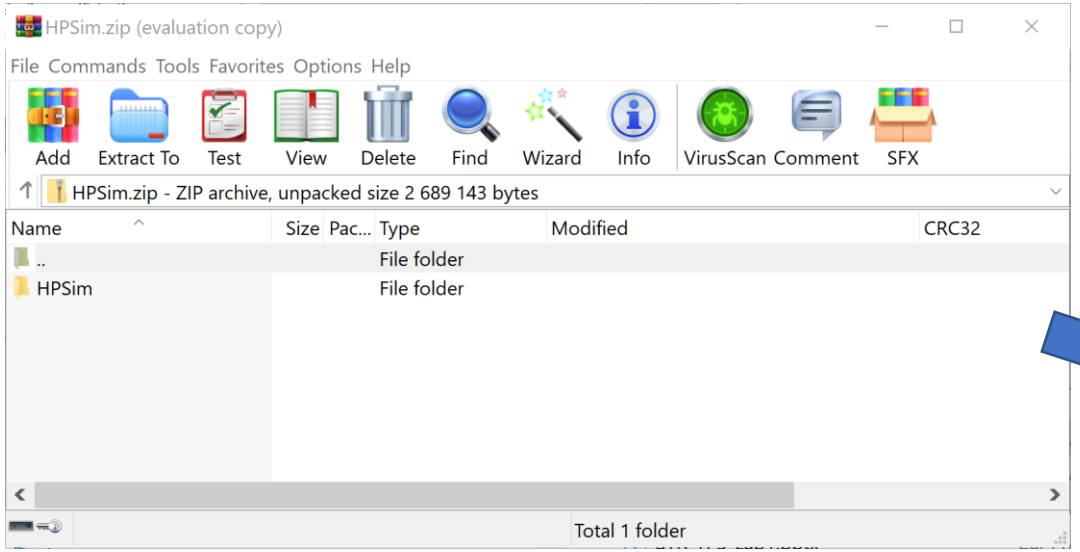
# REVISITING PETRI NET MODELING

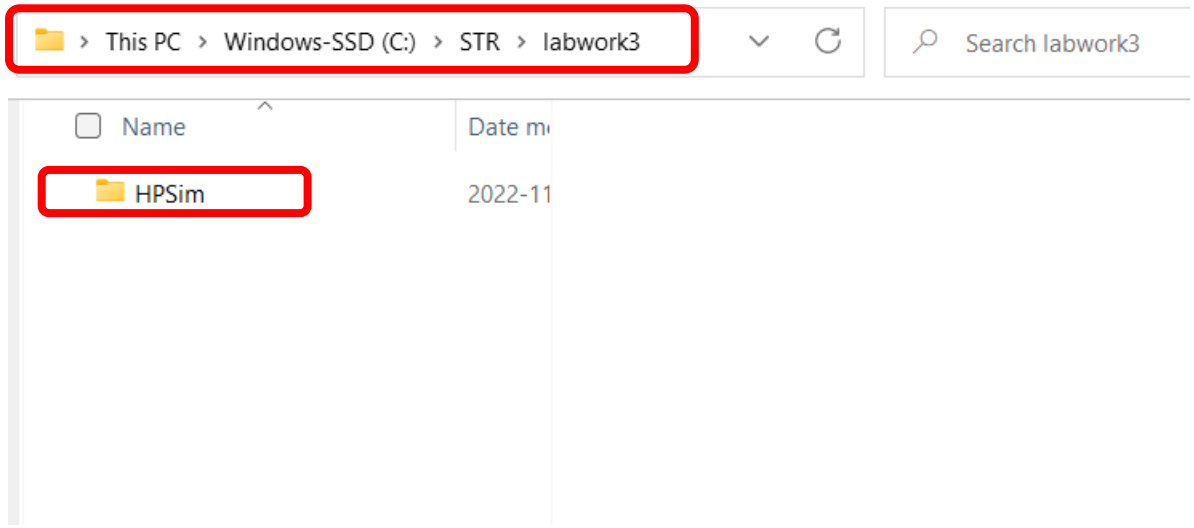
- Use theoretical slides for Petri net conceptualization.



# HPSim INSTALLATION

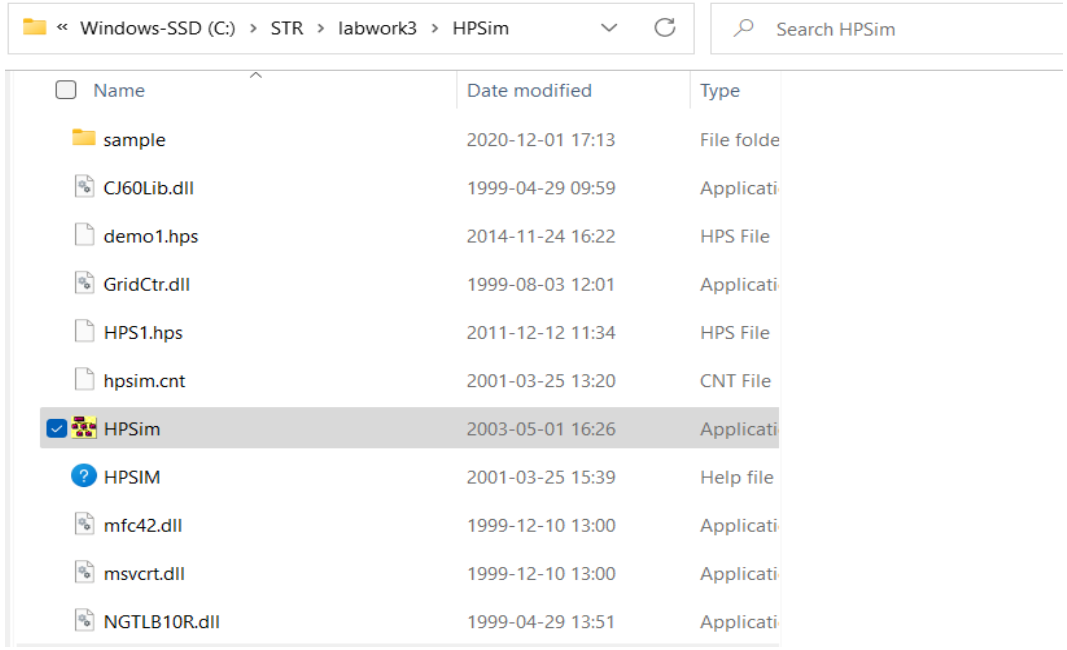
- Download the package **HPSim.zip** from CLIP
- Extract it inside **c:\str\labwork3** (strongly recommend it, posterior phases use absolute paths inside this folder)

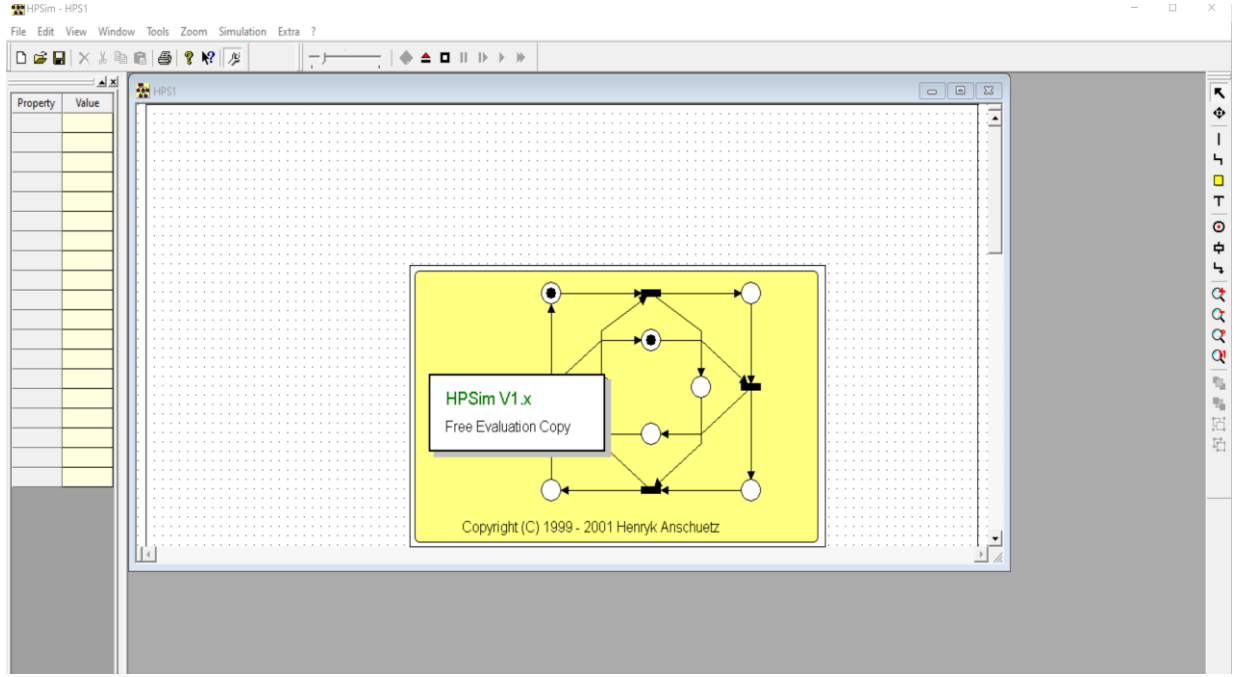


# RUNNING HPSim

- Although a bit old, the software has zero footprint on windows (no libraries spread inside system folders).
- To run it, just click in HPSim.exe.



# LEARNING SOFTWARE UTILIZATION

- To learn the software, we can rely on good external sources.

- See for instance, the manual available in CLIP and also in http://www.dee.ufrj.br/controle_automatico/cursos/Manual_HPSim.pdf (inside the document, the link for hp download is broken)

- A video (Spanish language) illustrating the HPSim capabilities is here: https://www.youtube.com/watch?v=DG2iEkuqcaU

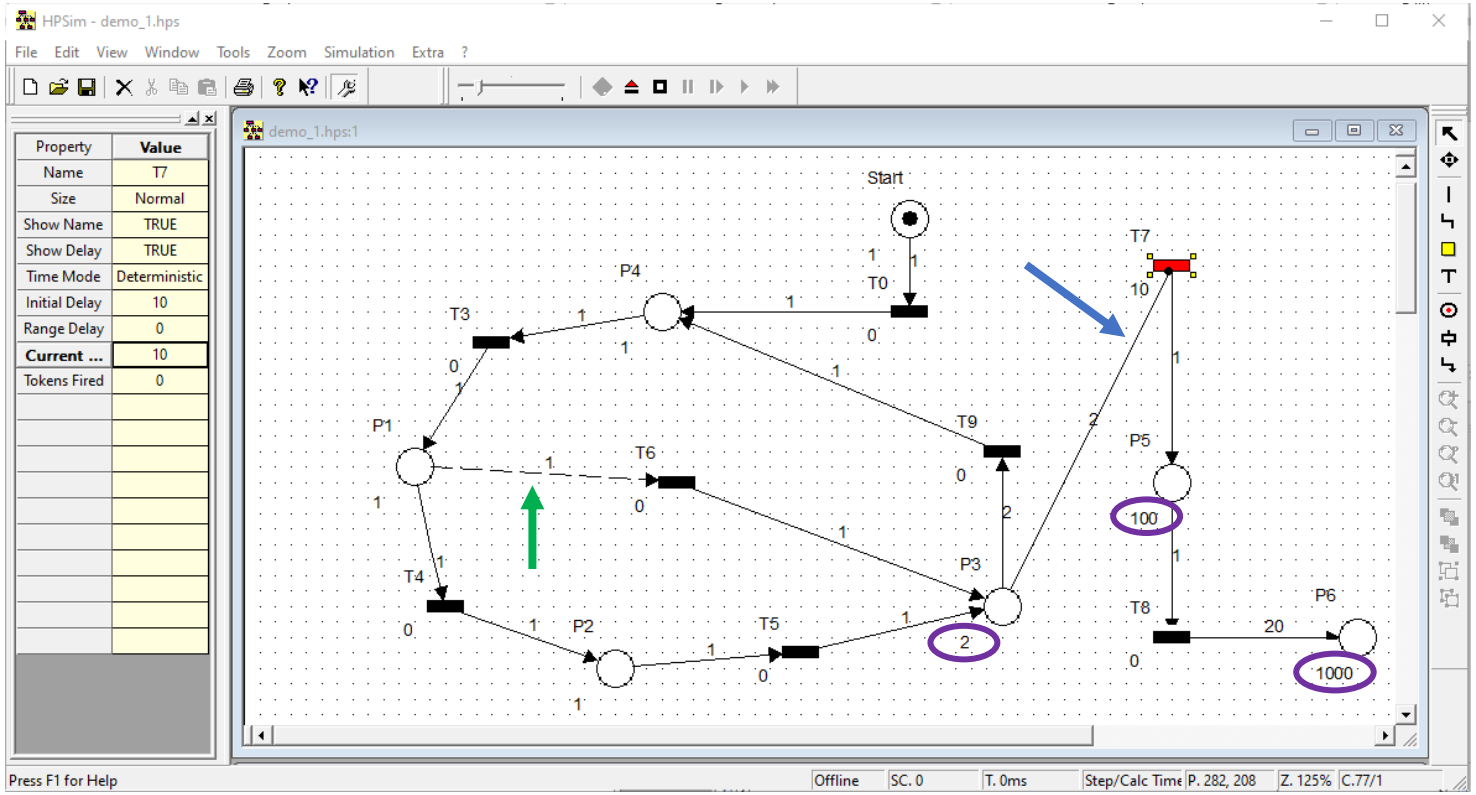
# RECOMMENDATION

- It is not frequent, but sometimes the software crashes.

- Therefore, while creating the Petri Net diagrams, save the work regularly.

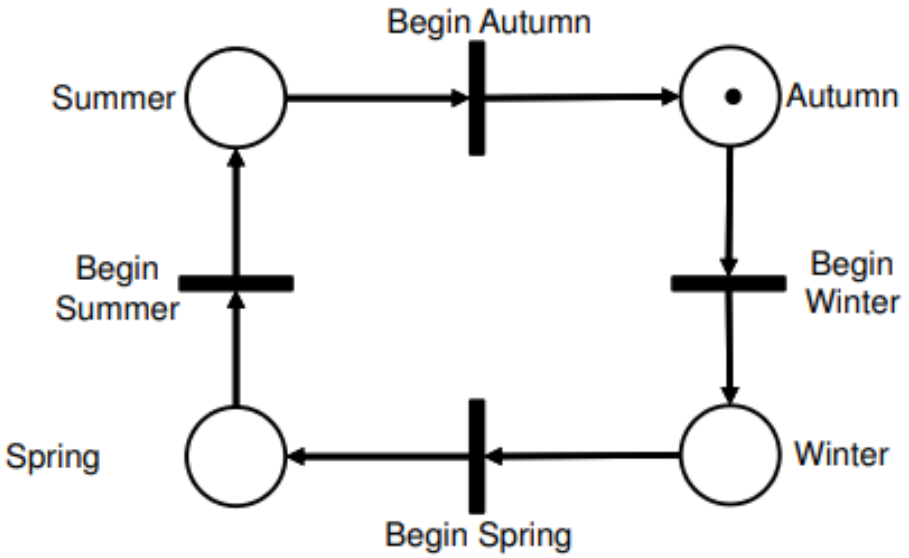
# ILLUSTRATIVE EXAMPLE

- The example shown in the picture illustrates a model containing:
  - Immediate and **timed** transitions
  - Places with variable **capacity**
  - **Normal**, **Testing** an **inhibitor** arcs
- Develop this model in your laptop and simulate to observe its behavior.



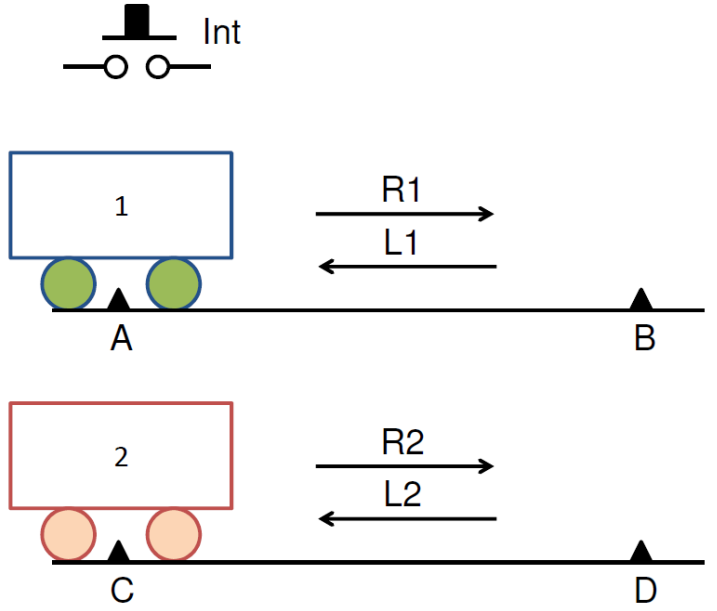
# TRAINING EXERCISE

- Model and observe the behavior of the petri-net shown in the picture.



# EXAMPLE FROM TP_9

- Two vehicles can move along two paths with different speeds.
- Closing Int, both vehicles start moving to the right.
- When a vehicle arrives as the end (B or D), it starts moving to the left, stopping in the initial position (A or C).
- The same cycle can be repeated whenever both vehicles are in the initial position and the switch Int is closed.

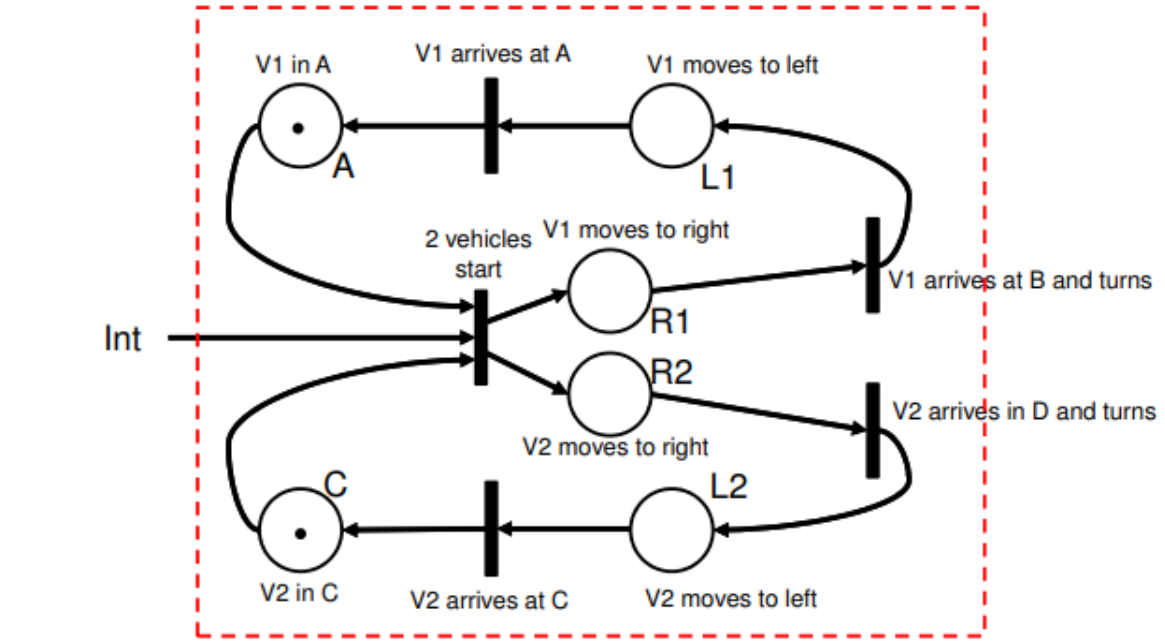



Represent this case in PN using HPSim.

# EXAMPLE FROM TP_9 – CONT.

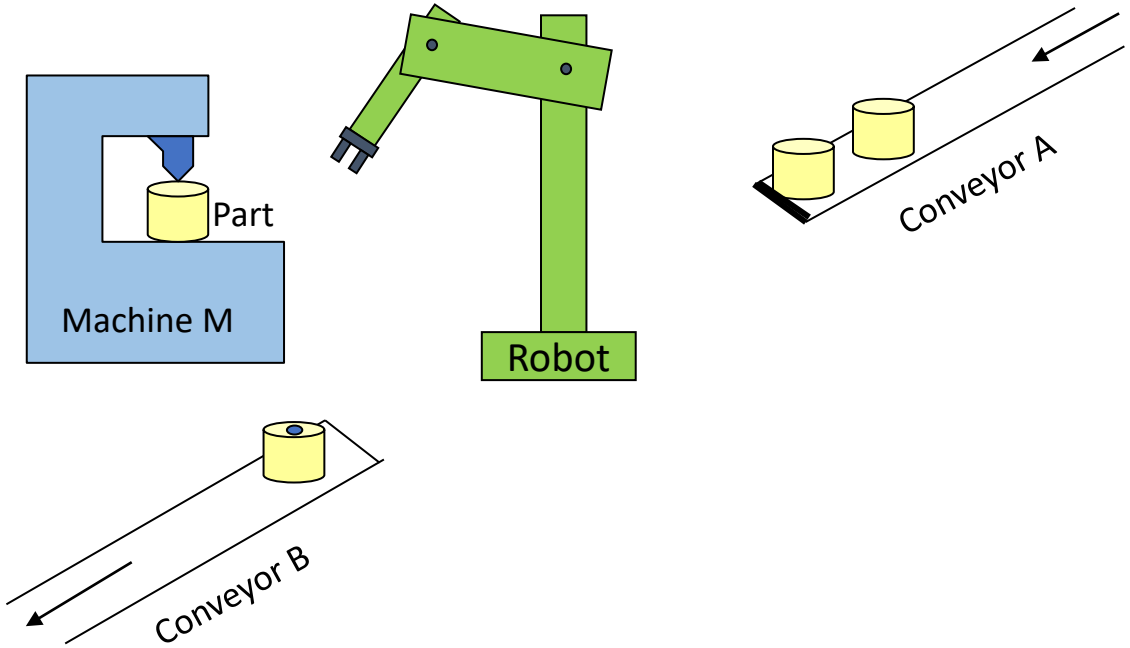
- Use a place with a token for modeling the “int” requests
- Try to conceive a solution for modeling more frequent requests

One solution:



# HPSim TO SOLVE HOMEWORK FROM TP_10

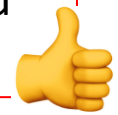
- Consider the following system. Parts to be processed arrive in conveyor A. The robot picks one part at a time from that conveyor and puts it in machine M for processing (when both the robot and the machine are free). When the machine finishes, the robot picks the processed part and puts it in the exit conveyor B.
- Describe the operation of this system using Petri nets. Define the initial marking corresponding to the situation shown in the figure.



# PROPOSED EXERCISES

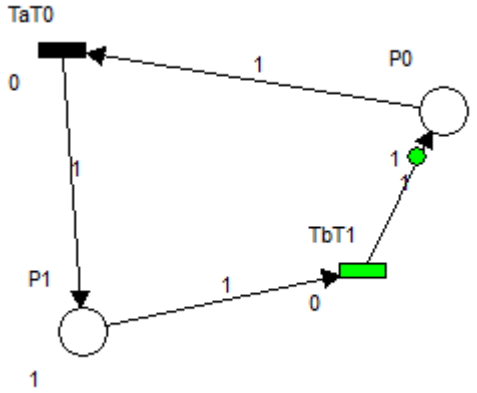
- We are now going to develop several models which allow training on the Petri Net concepts.
- Use the tables in the first section of the **report template (available on clip)** to provide answers.
- This report with your answers and the other requested information needs to be delivered with the implementation of labwork3.
- Use HPSim to develop the models for the following cases:
  1. An example of a safe PN, and an example of a bounded PN;
  2. An example of a conservative PN;
  3. A PN with capacity places and weighted arcs;
  4. A PN with capacity arcs of type “testing” and “inhibiting”.
  5. A PN with capacity arcs of type “testing” and “inhibiting”, places capacity and weighted arcs.
  6. A PN which starts working and enters in a deadlock state after a few iterations.

Feel free to use the examples of the theoretical classes and other sources

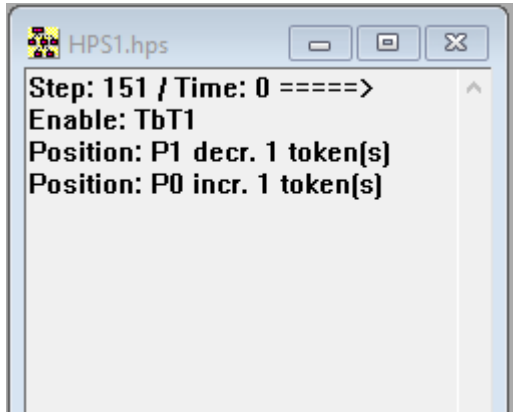


# EXAMPLE: MODELING A SECURE PN (DON'T USE THIS AS ANSWER)

- Copy/past the diagram from HPSim in this side and provide brief description of the used scenario/problem.




- Provide some explanations on its behavior during simulation in this side; copy/past the log window.

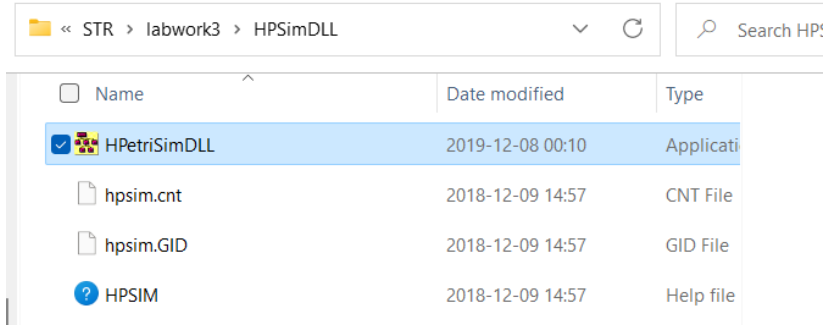


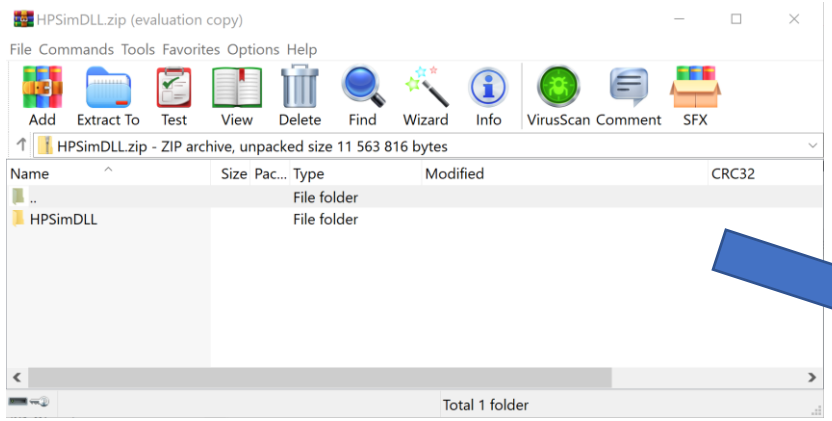
# Part 2

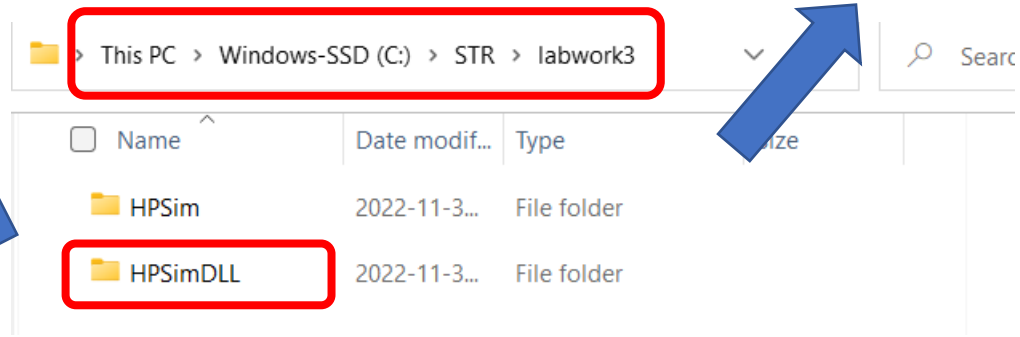
DLL integration in HPSim software

# HPSim NEW VERSION INSTALLATION

- Download the package HPSimDLL.zip from CLIP
- Extract it inside **c:\str\labwork3** (AGAIN strongly recommended)
- Launch the executable -> HPetriSimDLL.exe

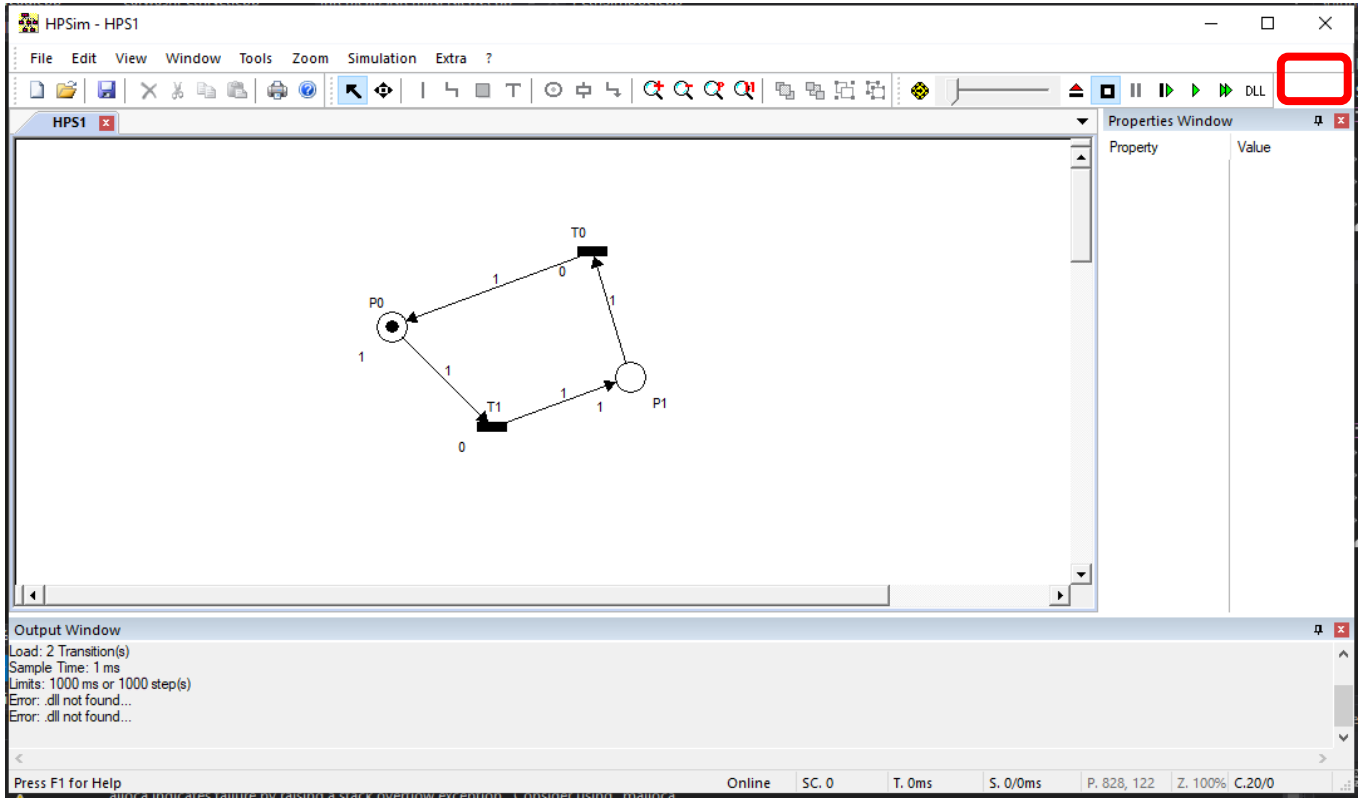


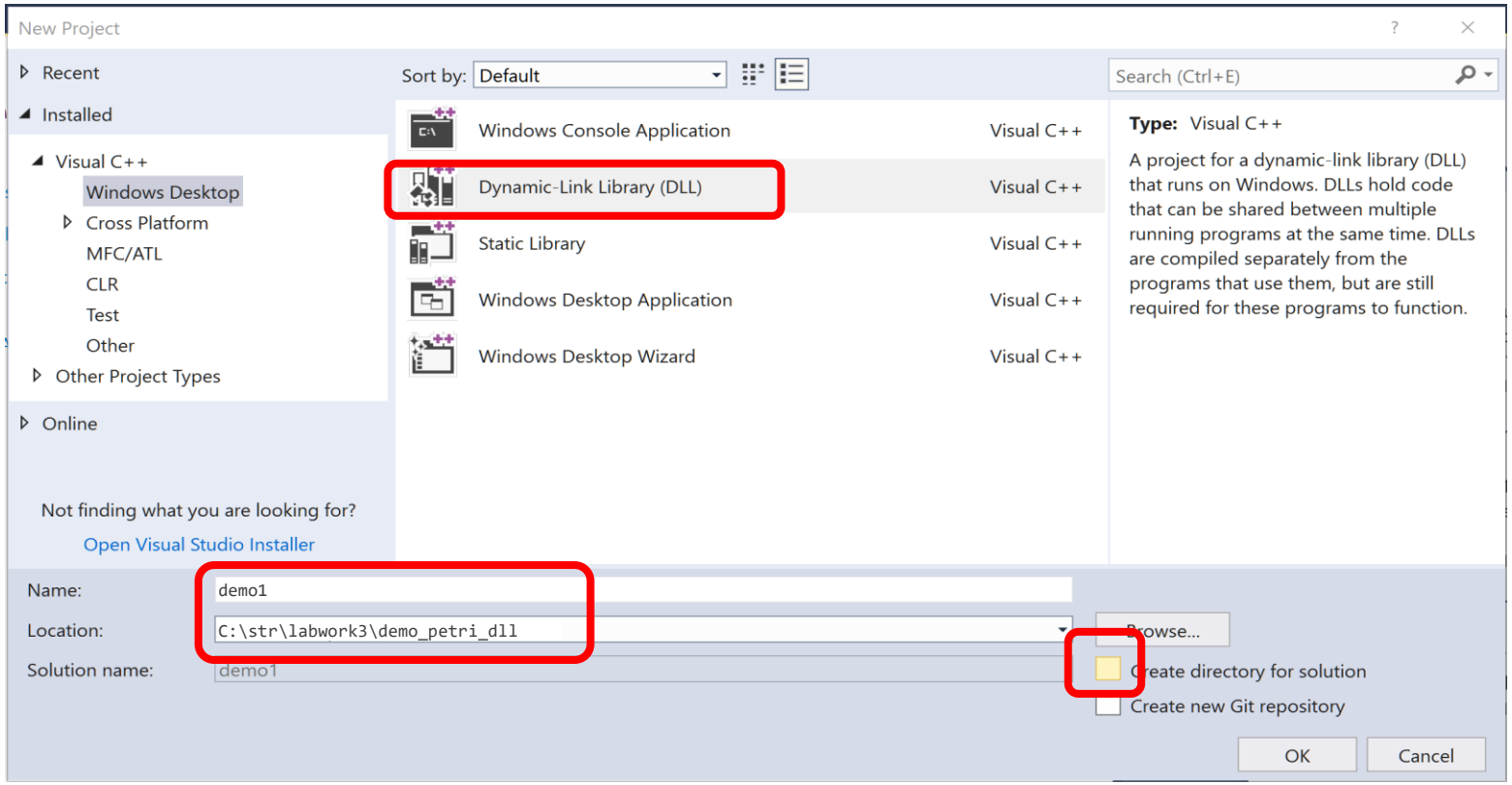
# HPSim NEW VERSION

- This version is pretty much the same.
- It has a new functionality which allows to associate a DLL to a Petri Net diagram.
- This way, HPSim can switch between PN simulator and PN executor.



# CREATING DLL (VS2017)

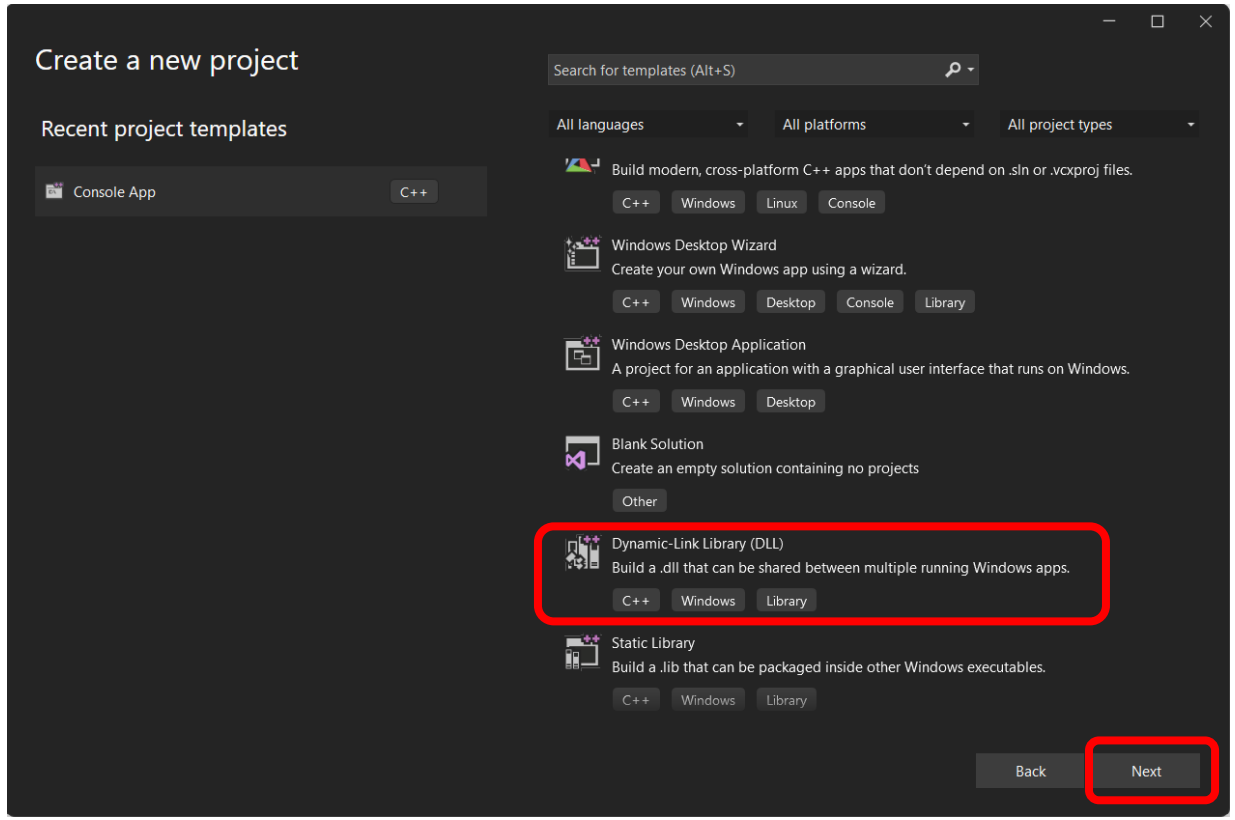
- Launch Visual Studio
- Select: Create New Project -> Windows Desktop -> Dynamic-Link Library

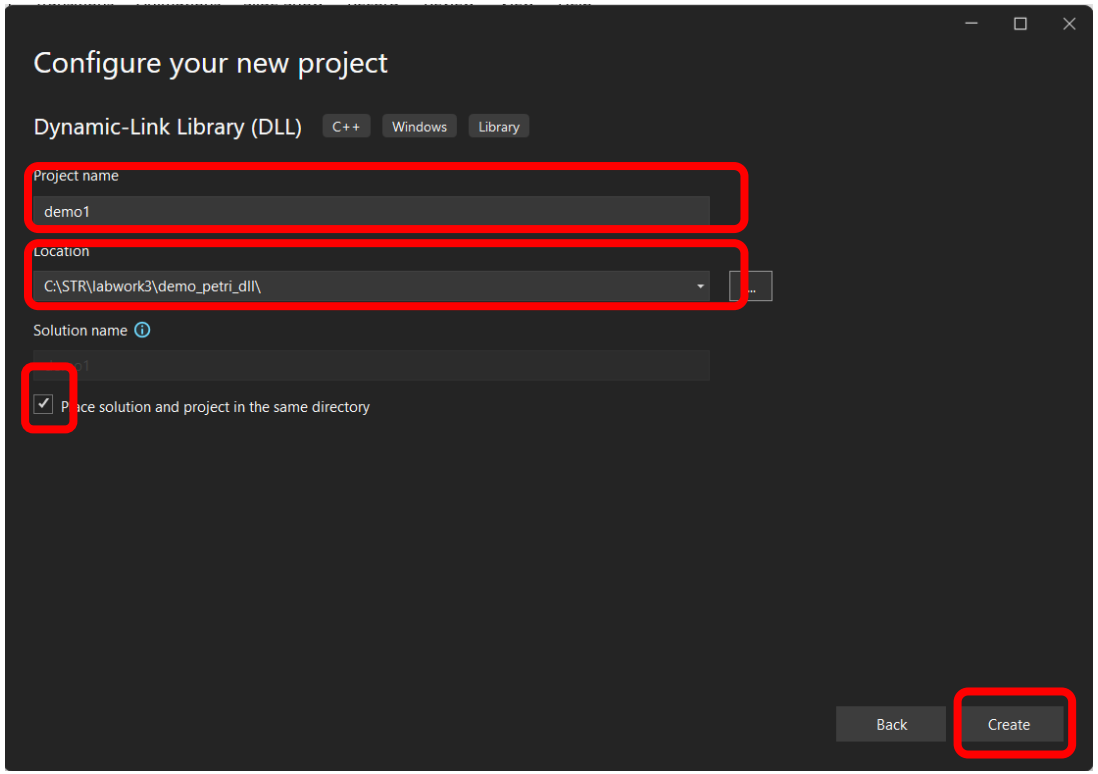



- Project Name: demo1
- Location: c:\str\labwork3\demo_petri_dll

# CREATING DLL (VS2019 / VS2022)

- Launch Visual Studio
- Select: Create a New Project -> Dynamic-Link Library (DLL)

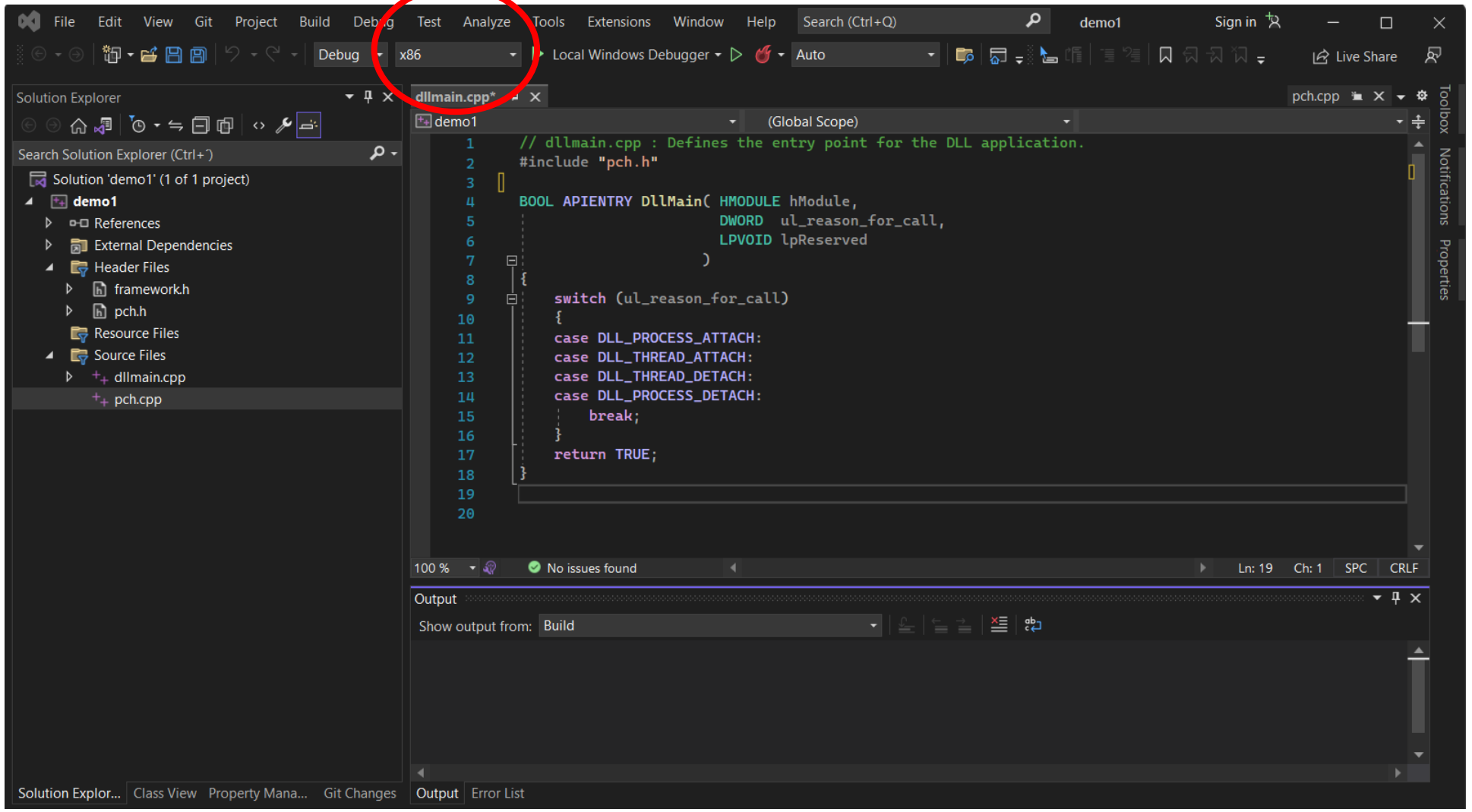


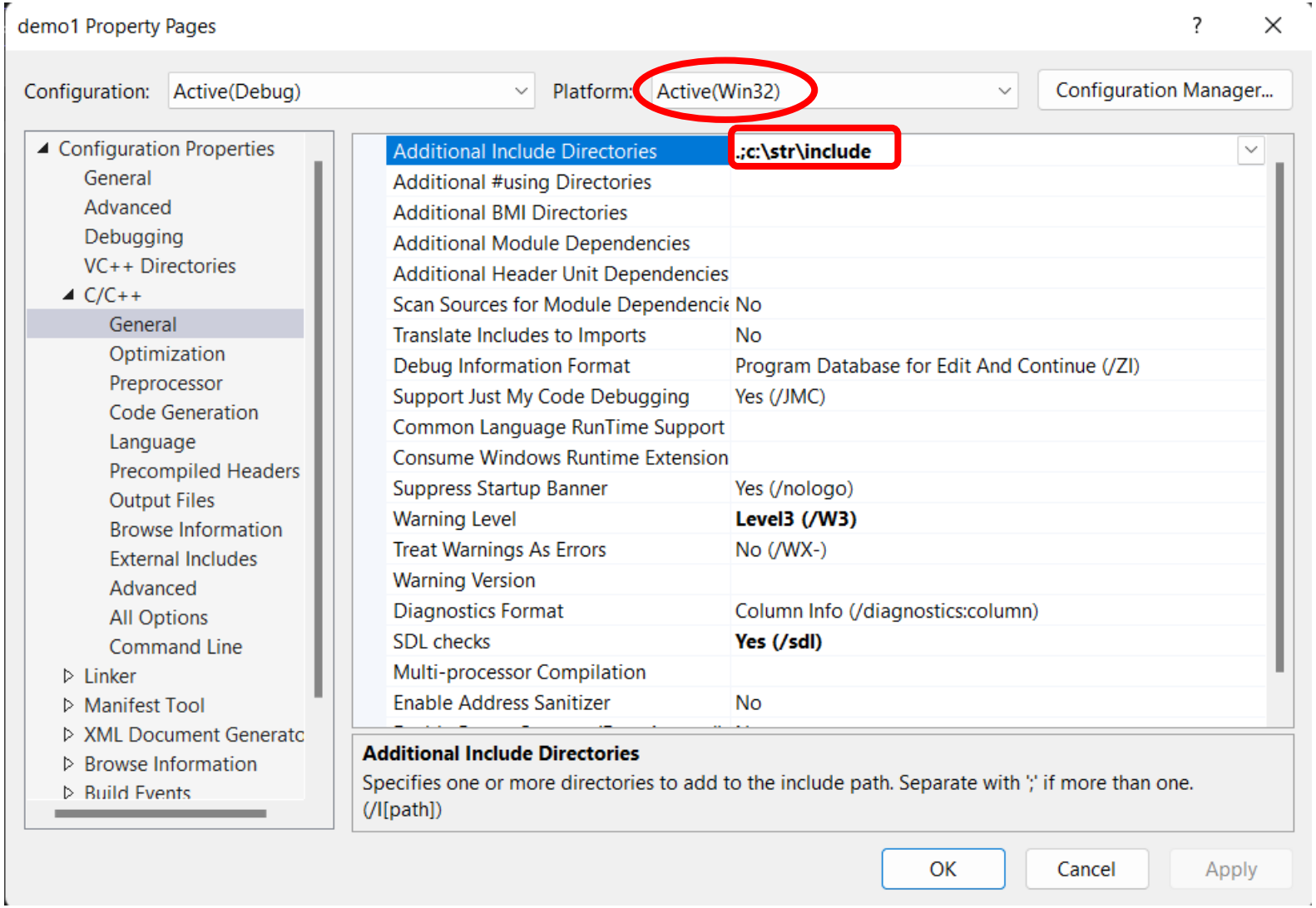

- Project Name: demo1
- Location: c:\str\labwork3\demo_petri_dll

# ASPECT OF CREATED DLL PROJECT

x86!!



# CONFIGURATION – “includes”

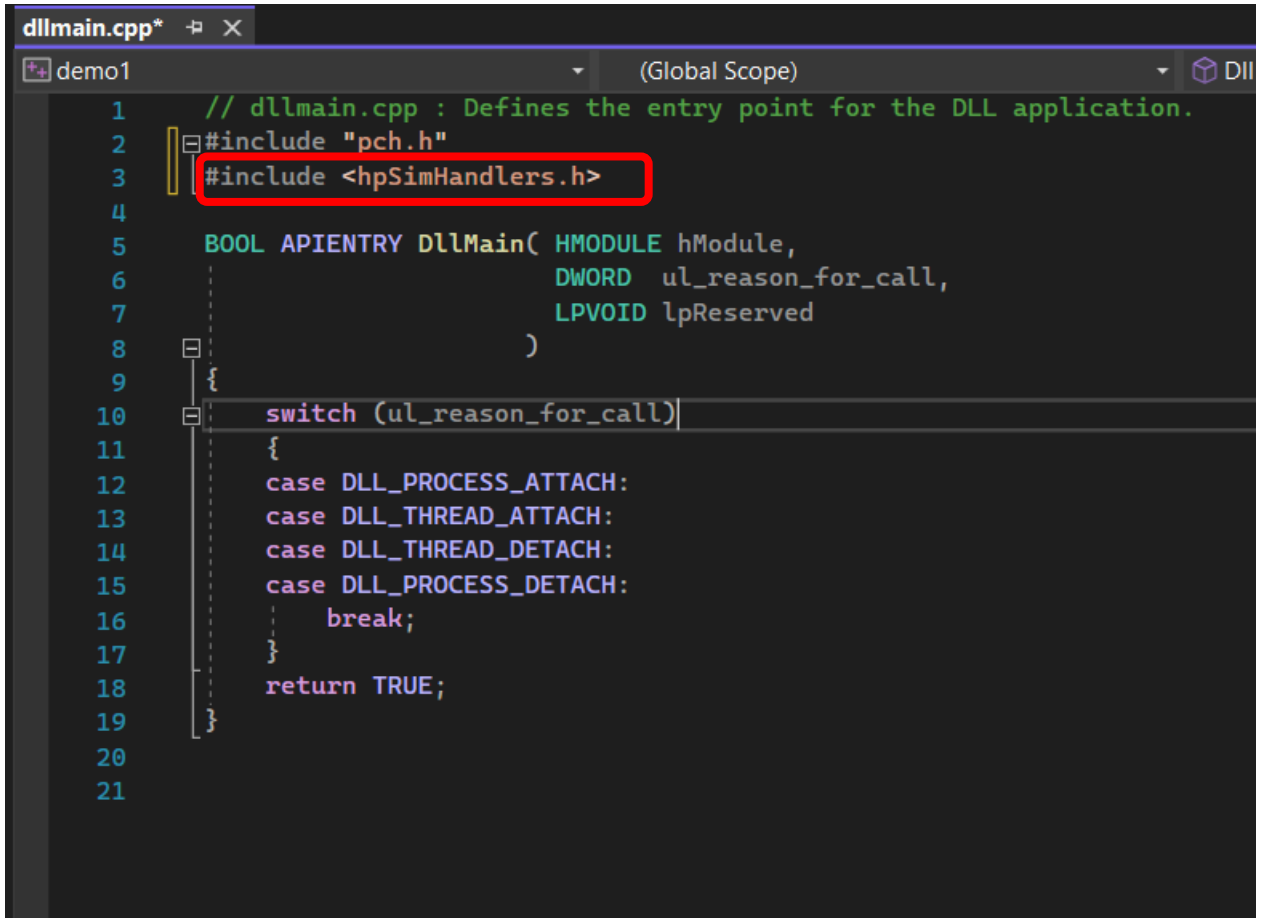



.;c:\str\include

# CONFIGURATION – “includes”

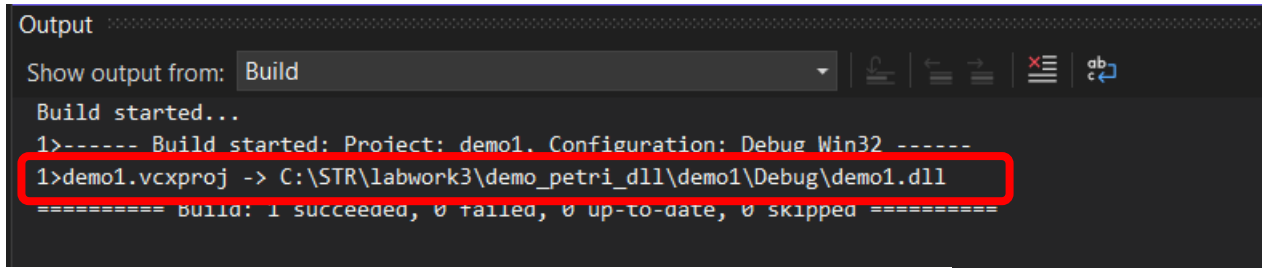
- Download the file **hpSimHandlers.h** from CLIP and save it into **c:\str\include**.

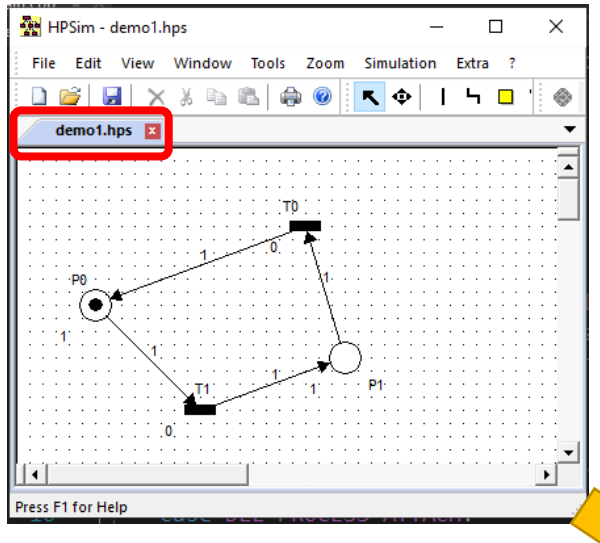
- Include it in **dllmain** source file and **build** the solution (not run!)

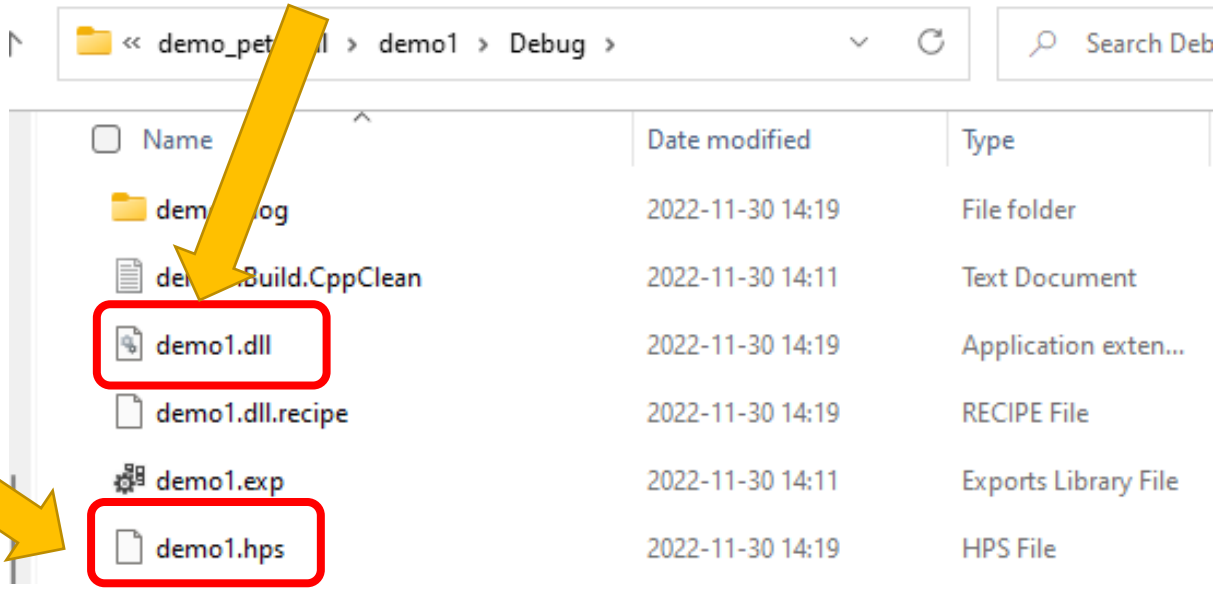


# INTEGRATION WITH PETRI NET

- Look at the output of the DLL in VStudio.



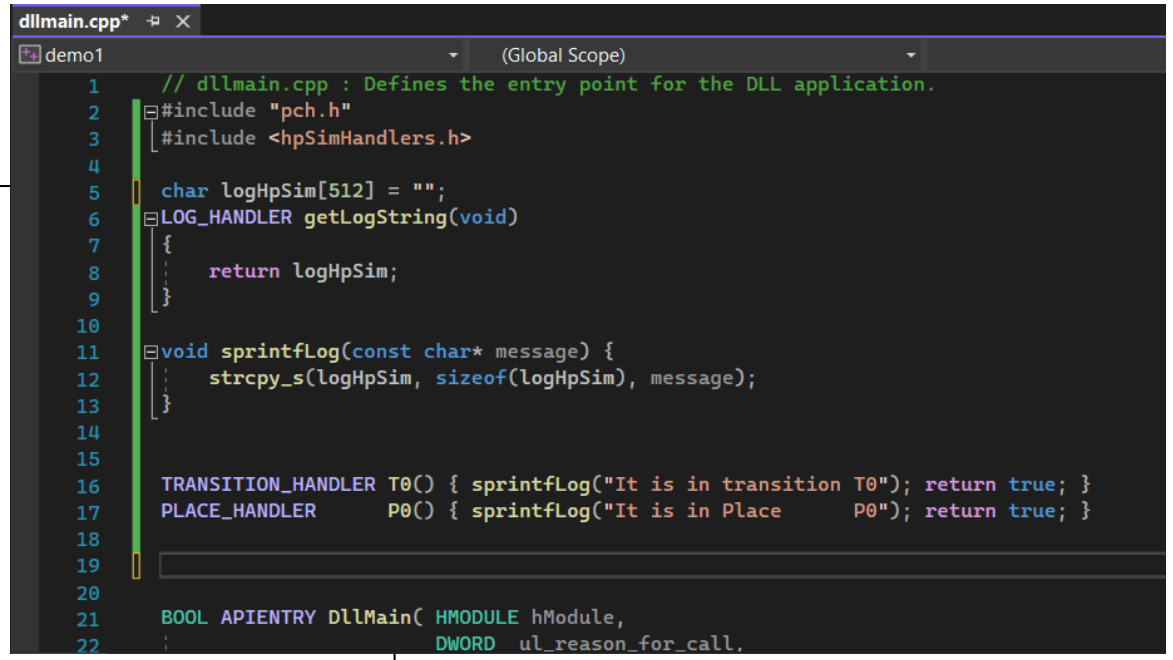




- **Save** the Petri Net file in the **same folder and same name** as the compiled DLL (IMPORTANT).

# INTEGRATION WITH PETRI NET (2)

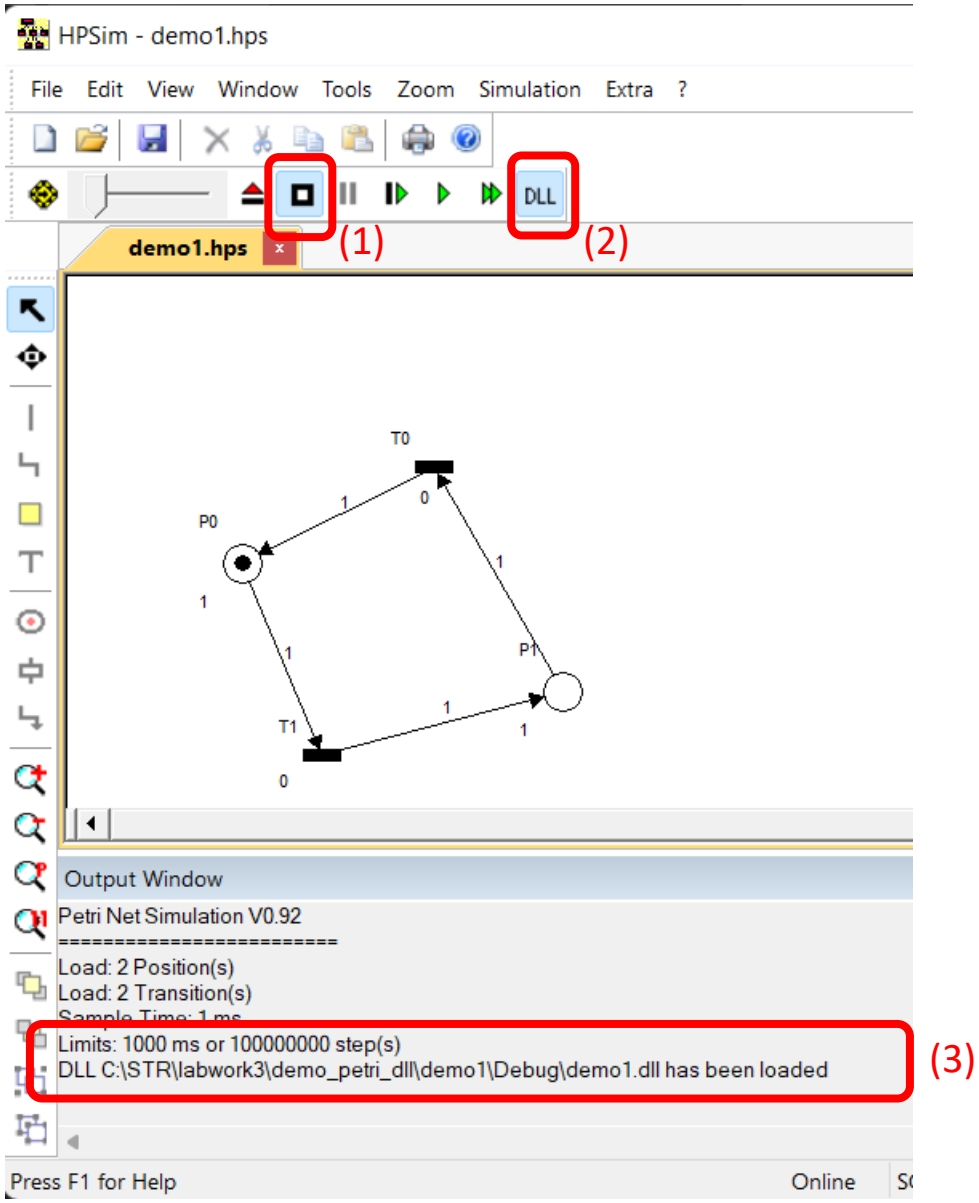
- Copy – paste the code below inside the dllmain.cpp (before `BOOL APIENTRY DllMain`)

```
#include "pch.h"
#include<hpSimHandlers.h>

char logHpSim[512] = "";
LOG_HANDLER getLogString(void)
{
   return logHpSim;
}

void sprintfLog(const char* message) {
strcpy_s(logHpSim, sizeof(logHpSim), message);
}


TRANSITION_HANDLER T0() { sprintfLog("It is in transition T0"); return true; }
PLACE_HANDLER      P0() { sprintfLog("It is in Place      P0"); return true; }
```



# PREPARING SIMULATION

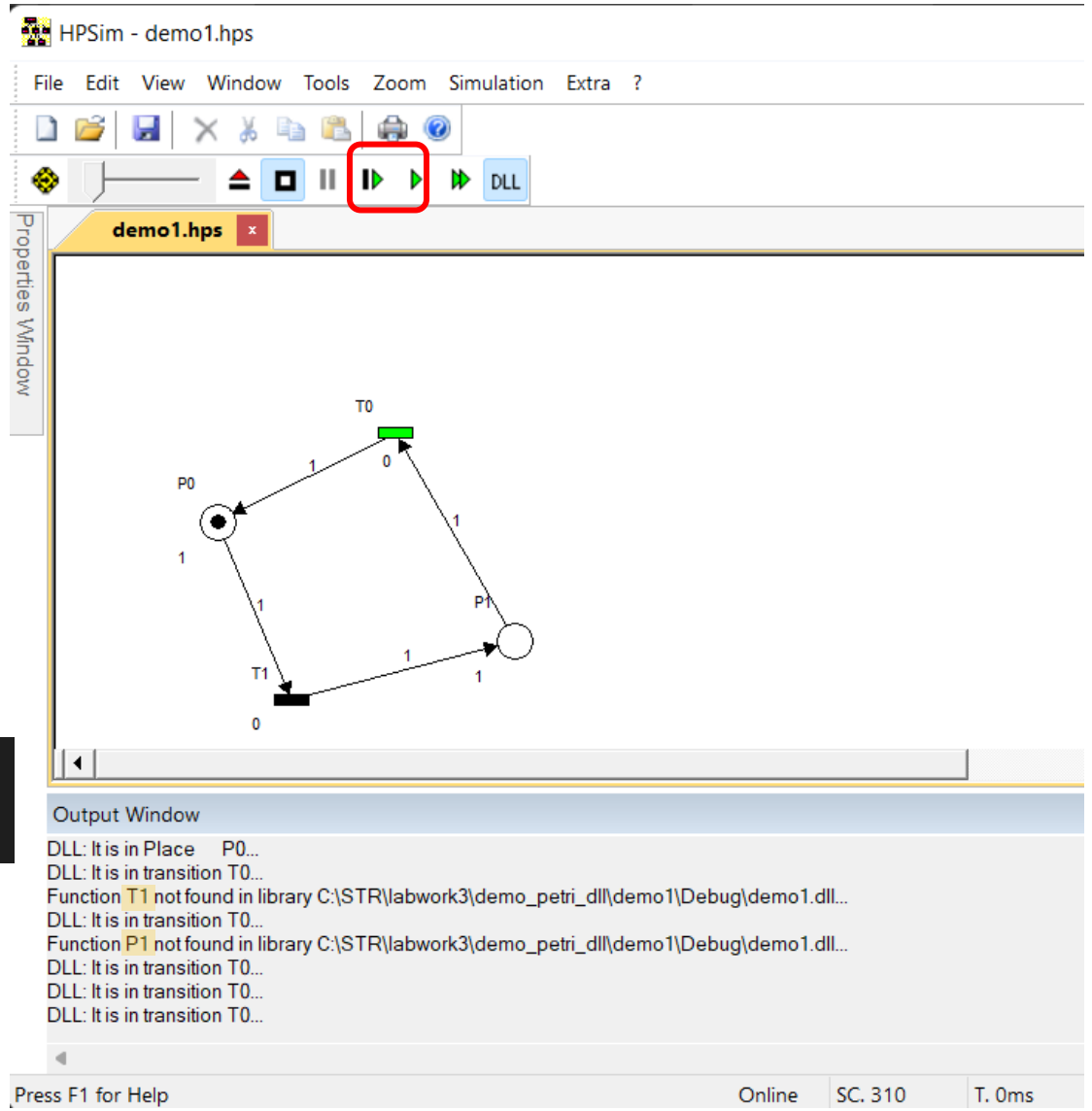
- Each time the DLL code is changed, it should be compiled; do it now.
- Now go into HPSim and get into simulation mode (1)
- Then, connect the model to the DLL by pressing DLL button (2).
- Notice the message in the Output Window telling "DLL:… dem1.dll has been loaded" (3)
- If not, ensure that both **demo1**.hps and **demo1**.dll have got the same name and in the same folder.



# RUNNING THE PETRI NET

- Now press the play button.
- Notice how the log messages from the handlers appear in the output window.
- Furthermore, transitions and places that were not assigned with a handler in the DLL are identified by corresponding alert message.

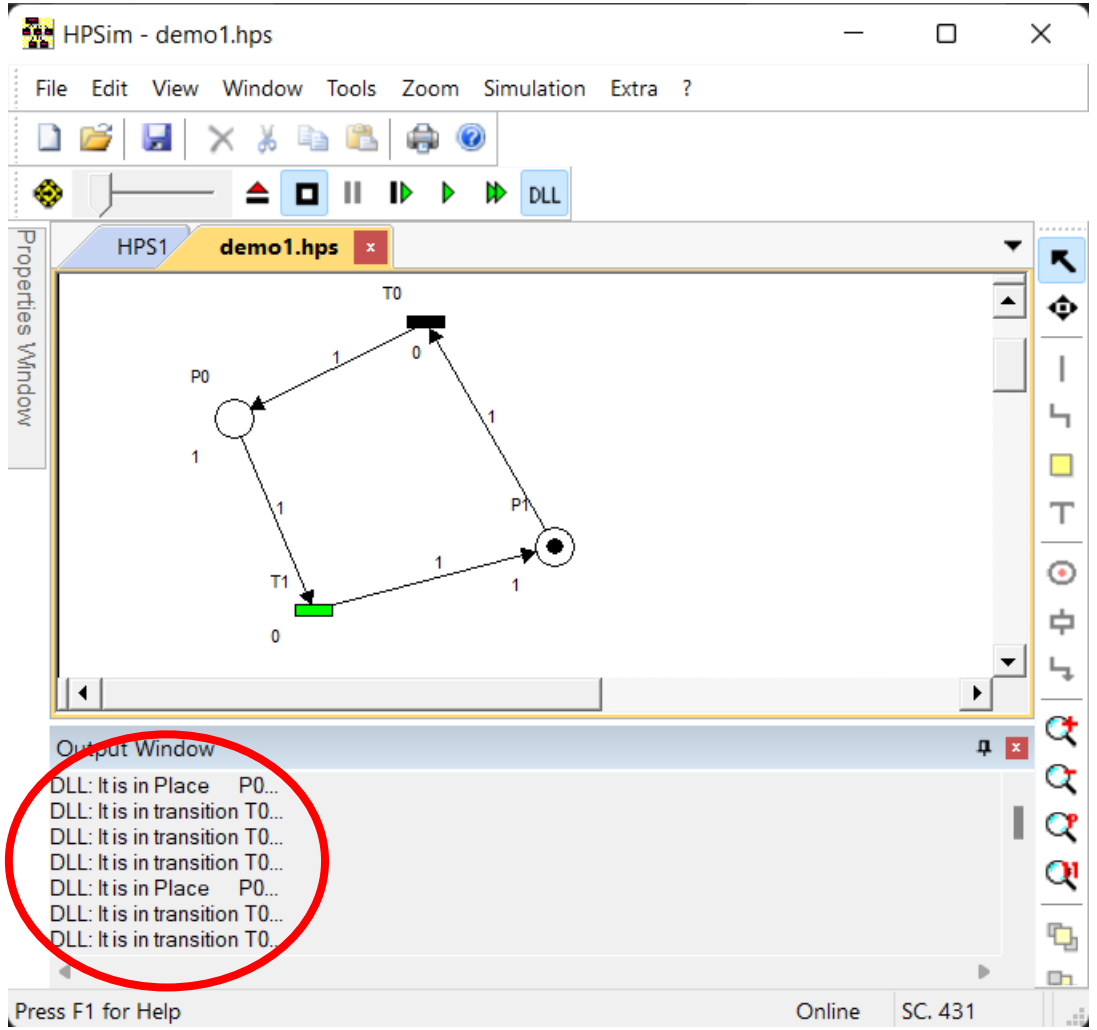
```
TRANSITION_HANDLER T0() { sprintfLog("It is in transition T0"); return true; }
PLACE_HANDLER      P0() { sprintfLog("It is in Place      P0"); return true; }
```



# DISABLING MISSING HANDLER MESSAGES

- In order to avoid such messages, you just need to define the functions shown below.
- As they return false, for now on, it is not necessary to define all the handlers.

```
LOG_ALERT_HANDLER alertMissingTransitions(void) {
    // returning false prevents the annoying messages
    return false;
}

LOG_ALERT_HANDLER alertMissingPlaces(void) {
    // returning false prevents the annoying messages
    return false;
}
```



# RULES FOR FIRING TRANSITIONS

- Beyond the usual petri net rules for firing transitions, we can use Boolean conditions, based on e.g., sensors.
- In such a way, handlers of transitions will encode these conditions. A transition would fire if all places connected to it have the necessary tokens and Boolean condition yields true:

```
TRANSITION_HANDLER T0() {
    sprintfLog("It is in transition T0");
    uInt8 p = readDigitalU8(1);
    bool partDetected = (bool)getBitValue(p, 3);
    return partDetected;
}
```

# ASSOCIATING ACTIONS TO A PLACE

- We can associate actions in C++ to the places in a PN
- Whenever a place has a token, the corresponding action is executed.
- When a handler of a place returns true, the action is executed just once
- When a handler of a place returns false, the action is executed in every step of the execution.

```
PLACE_HANDLER P0() {
    sprintfLog("It is in Place      P0");
    stopVerticalMovement();
    moveBrushRight();
    return true; // execute only once
}
```

# OTHER HANDLERS

- Other handlers can be associated to Places; they are:

```
TOKEN_CHANGED_HANDLER onTokenEnterPlace_P0(long tokensBefore, long tokensNow) {
    if (tokensBefore == 0 && tokensNow > 0) {
        //do something here
    }
}

TOKEN_CHANGED_HANDLER onTokenLeavePlace_moveDown(long tokensBefore, long tokensNow) {
    if (tokensBefore >= 0 && tokensNow == 0) {
        //do something here
    }
}
```

- These handlers are executed only once, just when a token enters or leaves a Place.

# NEXT LAB WORK

- Applying the Petri Net modeling in the solution of a Car Wash Services system
- Installation of CarWash simulator
- Project interaction with HPSim
- User interface

# KEEP UP THE GOOD WORK!